

Linear Equations

Geometric Algorithms
Lecture 1

Outline

- » Motivating examples of linear systems
- » Formally define linear systems
- » Solve systems of linear equations

Keywords

linear equations

coefficient

unknowns

point sets

hyperplanes

systems of linear equations

consistency

augmented matrix

coefficient matrix

forward-elimination

back-substitution

row operations

row equivalence

Motivation

Lines (Slope-Intercept Form)

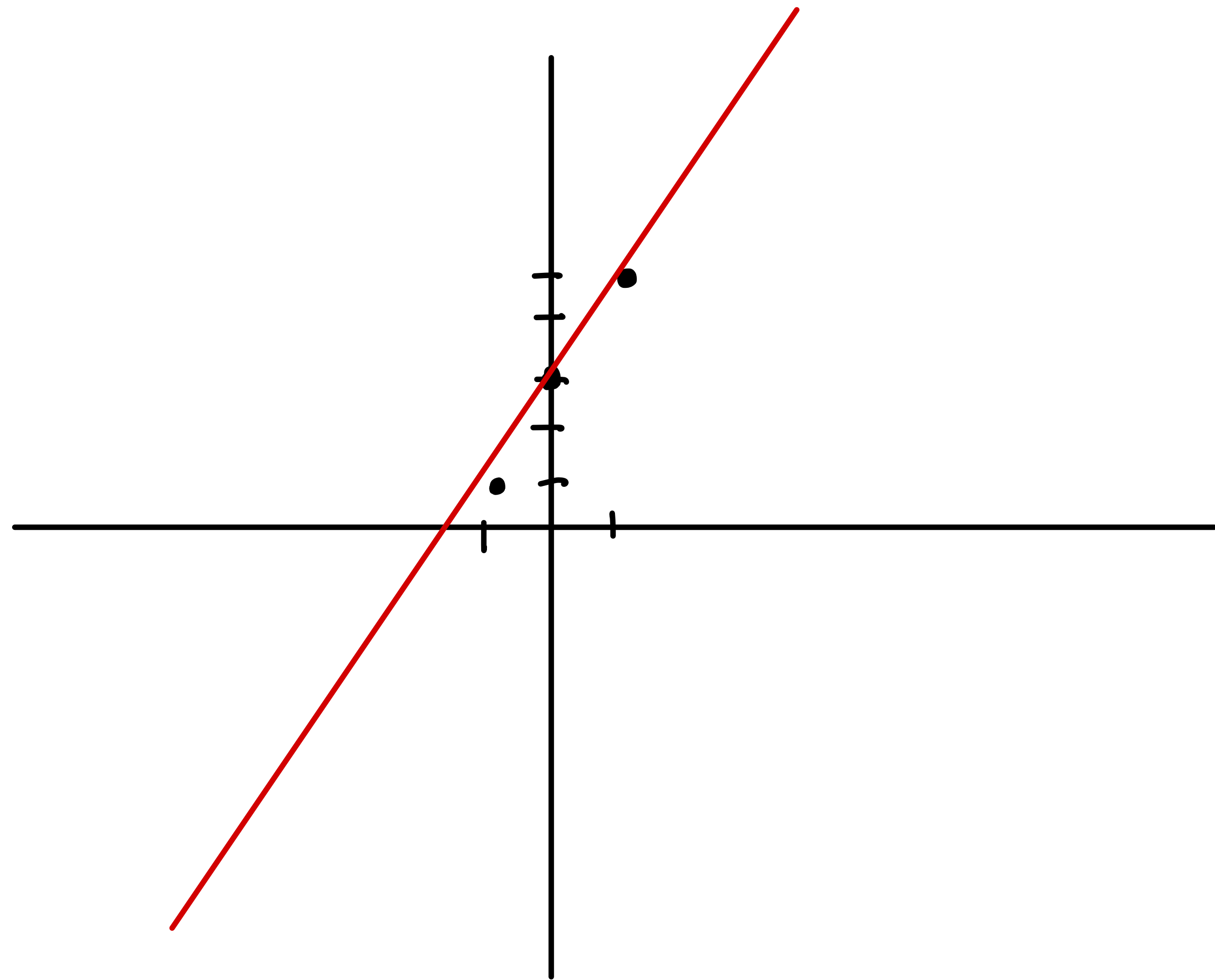
$$y = \underbrace{m}_\text{slope} x + \underbrace{b}_{y\text{-int.}}$$

$$y = 2x + 3$$

$$0 \mapsto 3$$

$$1 \mapsto 5$$

$$-1 \mapsto 1$$



Lines (Slope-Intercept Form)

$$y = mx + b$$

Lines (Slope-Intercept Form)

$$y = \textcolor{red}{mx} + \textcolor{blue}{b}$$

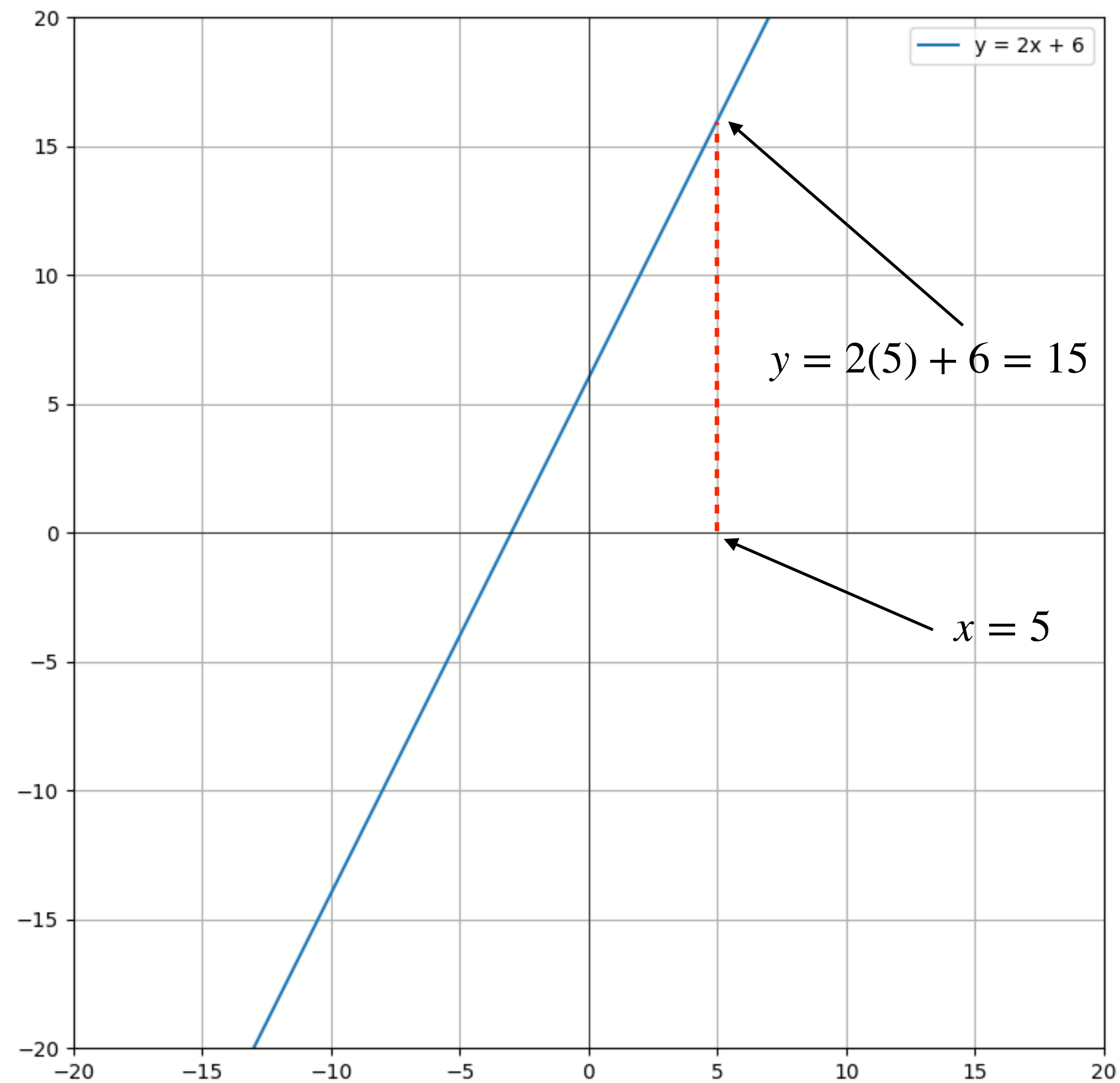
slope y-intercept

Lines (Slope-Intercept Form)

$$y = \underbrace{mx}_{\text{slope}} + \underbrace{b}_{\text{y-intercept}}$$

Given a value of x , compute a value of y

Lines (Graph)



Lines (General Form)

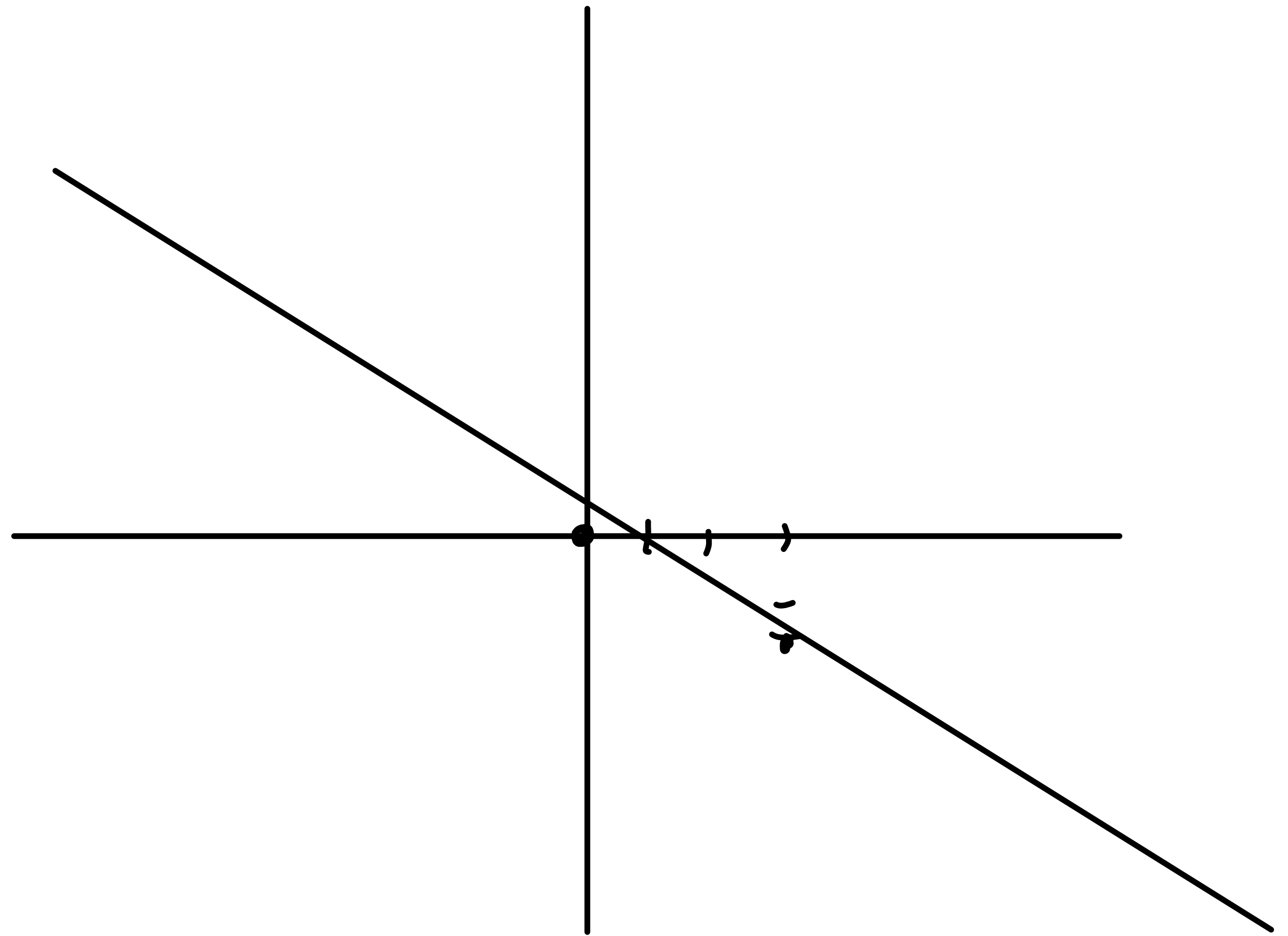
$$a x + b y = c$$

$$2x + 3y = 0$$

$$(0, 0)$$

$$(6, -4)$$

$$(3, -2)$$



Lines (General Form)

$$ax + by = c$$

Lines (General Form)

$$ax + by = c$$

x-intercept: $-\frac{c}{a}$

Lines (General Form)

$$ax + by = c$$

x-intercept: $\frac{c}{a}$

y-intercept: $\frac{c}{b}$

Lines (General Form)

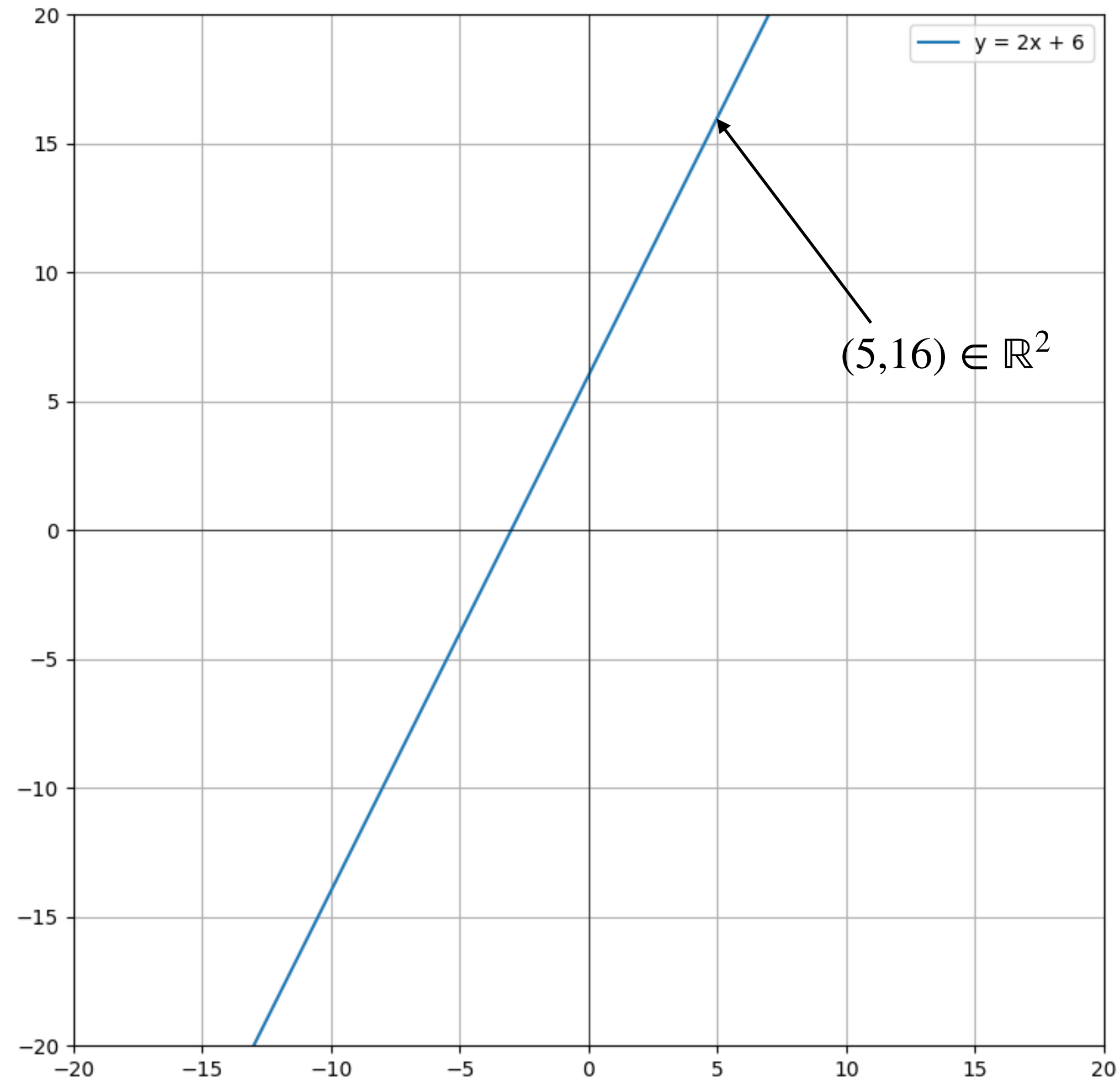
$$ax + by = c$$

$$\text{x-intercept: } \frac{c}{a}$$

$$\text{y-intercept: } \frac{c}{b}$$

What values of x and y make the equality hold?

Lines (Graph)



$$\{(x, y) : (-2)x + y = 6\}$$

Lines

slope-int $\rightarrow$ general

$$(-m)x + y = b$$

general $\rightarrow$ slope-int

$$y = \left(\frac{-a}{b} \right) x + \frac{c}{b}$$

Line Intersection

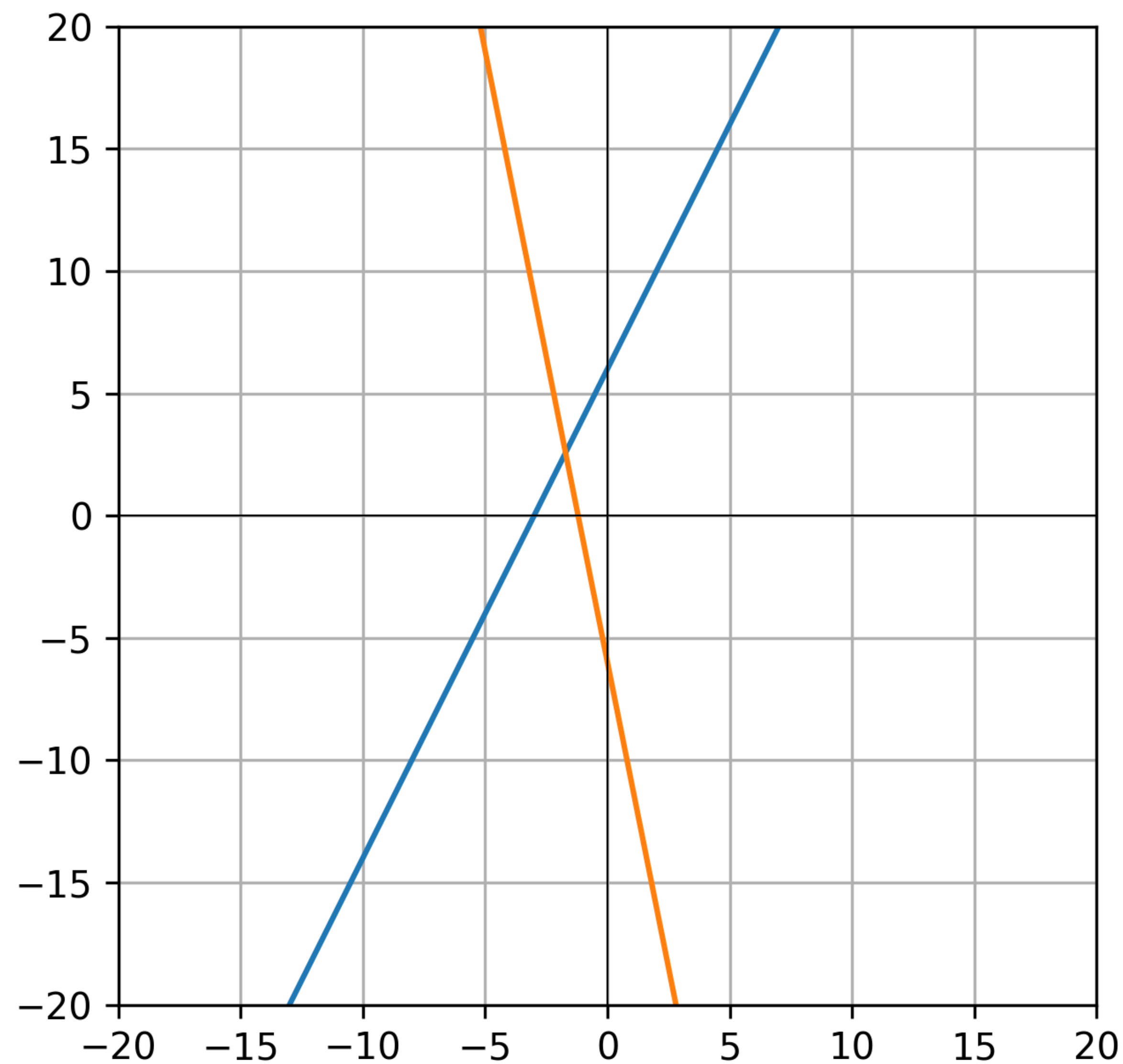
$$y = m_1x + b_1$$

$$y = m_2x + b_2$$

Q. Given two lines, where do they intersect?

Line Intersection

Line Intersection (Graph)



Line Intersection (Alternative)

Line Intersection (Alternative)

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Q. Given two (general form) lines, what values of x and y satisfy ***both*** equations?

Line Intersection (Alternative)

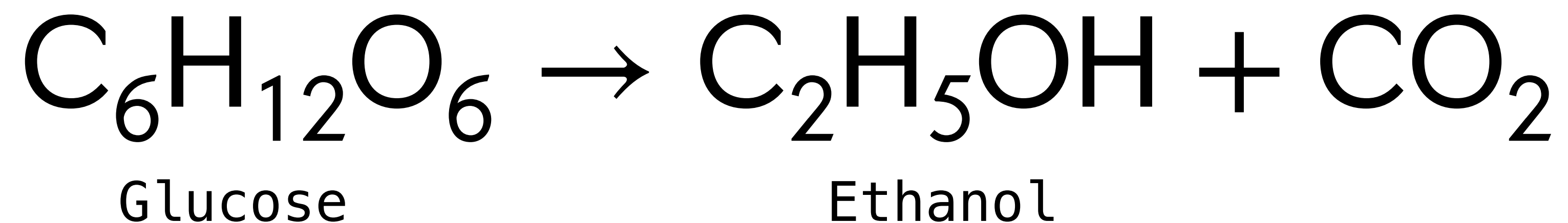
$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

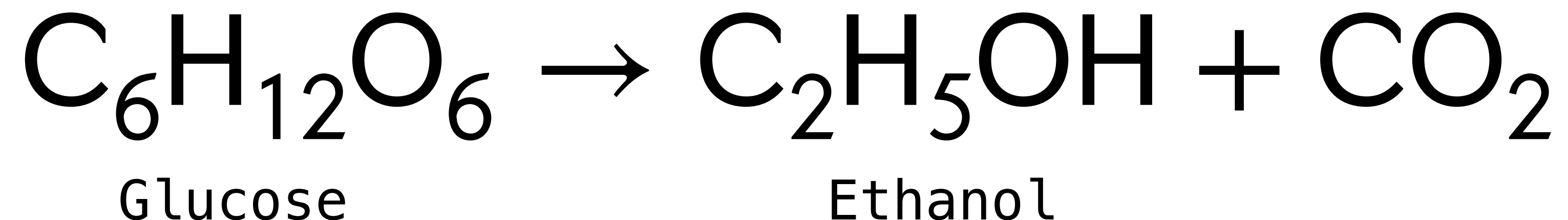
Q. Given two (general form) lines, what values of x and y satisfy ***both*** equations?

This is the same question

Example: Balancing Chemical Equations

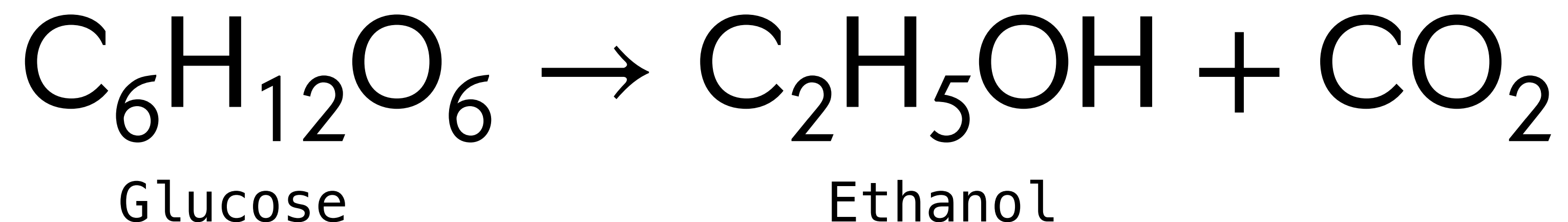


Example: Balancing Chemical Equations



How much ethanol is produced by fermentation?
(for science)

Example: Balancing Chemical Equations



How much ethanol is produced by fermentation?
(for science)

The **number of atoms** has to be *preserved* on each side of the equation

Balancing Chemical Equations



Balancing Chemical Equations



$$6\alpha = 2\beta + \gamma \quad (\text{C})$$

$$12\alpha = 6\beta \quad (\text{H})$$

$$6\alpha = \beta + 2\gamma \quad (\text{O})$$

Balancing Chemical Equations



$$6\alpha - 2\beta - \gamma = 0 \quad (\text{C})$$

$$12\alpha - 6\beta = 0 \quad (\text{H})$$

$$6\alpha - \beta - 2\gamma = 0 \quad (\text{O})$$

Formal Definitions

Linear Equations

$$2x + 3y + 4z + 5w = 7 \quad \checkmark$$

$$\pi x + ey = 4 \quad \checkmark$$

$$3x = 2y + z \quad \checkmark$$

$$2x^2 = 3 \quad \times$$

$$2xy + zw = 5 \quad \times$$

$$\log x + 2^x = 4 \quad \times$$

Linear Equations

Df. A *linear equation* in variables $x_1, x_2, \dots, x_n$ is an equation which can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n, b$ are real numbers ($\mathbb{R}$)

Linear Equations

Df. A *linear equation* in variables $x_1, x_2, \dots, x_n$ is an equation which can be written in the form

coefficients

$$\boxed{a_1}x_1 + \boxed{a_2}x_2 + \dots + \boxed{a_n}x_n = b$$

where $a_1, a_2, \dots, a_n, b$ are real numbers ($\mathbb{R}$)

Linear Equations

Df. A *linear equation* in variables $x_1, x_2, \dots, x_n$ is an equation which can be written in the form

$$\begin{array}{c} \text{coefficients} \qquad \qquad \qquad \text{unknowns} \\ a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b \end{array}$$

where $a_1, a_2, \dots, a_n, b$ are real numbers ($\mathbb{R}$)

Point sets

Point sets

Linear equations describe *point sets*:

$$\{(s_1, s_2, \dots, s_n) \in \mathbb{R}^n : a_1 s_1 + a_2 s_2 + \dots + a_n s_n = b\}$$

Point sets

Linear equations describe *point sets*:

Euclidean Space

$$\{(s_1, s_2, \dots, s_n) \in \mathbb{R}^n : a_1s_1 + a_2s_2 + \dots + a_ns_n = b\}$$

The points such that the equation holds

Linear Equations, Geometrically

If a 2D linear equation represents a *line* in the plane, then a 3D linear equation is...

Linear Equations, Geometrically

If a 2D linear equation represents a *line* in the plane, then a 3D linear equation is...

Not a line...

Demo

$$z = 5$$

$$0x + 0y + z = 5$$

What does the point set of this linear equation look like?

$$\{ (x, y, z) \in \mathbb{R}^3 : z = 5 \}$$

$$(0, 0, 5) \quad (2, 2, 5)$$

$$(1, 1, 5) \quad (\pi, -e, 5)$$

Demo

$$-x + 0y + z = 5$$

How about this one?

$$\{ (x, y, z) \in \mathbb{R}^3 : z = x + 5 \}$$

$$(0, 5, \boxed{5})$$

$$(0, 0, 5)$$

$$(0, \pi, 5)$$

$$(1, 0, \boxed{6})$$

Demo

$$-x + -y + z = 5$$

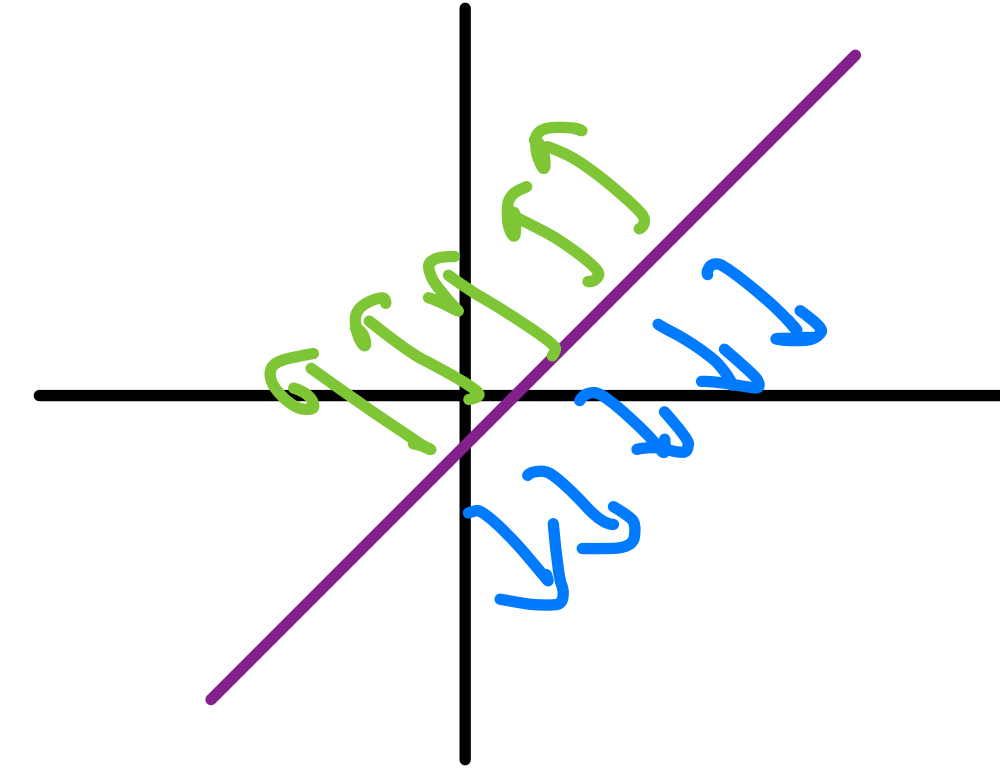
How about this one?

Hyperplanes

Hyperplanes

The point set of a general linear equation
is called a **hyperplane**

Hyperplanes



The point set of a general linear equation is called a **hyperplane**

Theme of the course: Hyperplanes "behave" like lines and planes in many respects

Linear Equations, Geometrically

If a 2D linear equation represents a *line* in the plane, then a 3D linear equation is...

Linear Equations, Geometrically

If a 2D linear equation represents a *line* in the plane, then a 3D linear equation is...

Not a line...

Linear Equations, Geometrically

If a 2D linear equation represents a *line* in the plane, then a 3D linear equation is...

A plane!

Systems of Linear Equations

Df. A *system of linear equations* is just a collection of linear equations *over the same variables*

Systems of Linear Equations

Df. A *system of linear equations* is just a collection of linear equations *over the same variables*

Df. A *solution* to a system is a point that satisfies all its equations *simultaneously*

Example

$$x + 2y = 1$$

$$-x - y - z = -1$$

$$2x + 6y - z = 1$$

Show that $(3, -1, -1)$ is solution to the above linear system

$$3 + 2(-1) = 1 \checkmark$$

$$-3 - (-1) - (-1) = -1 \checkmark$$

$$2(3) + 6(-1) - (-1) = 1 \checkmark$$

System of Linear equations (General-form)

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

System of Linear equations (General-form)

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

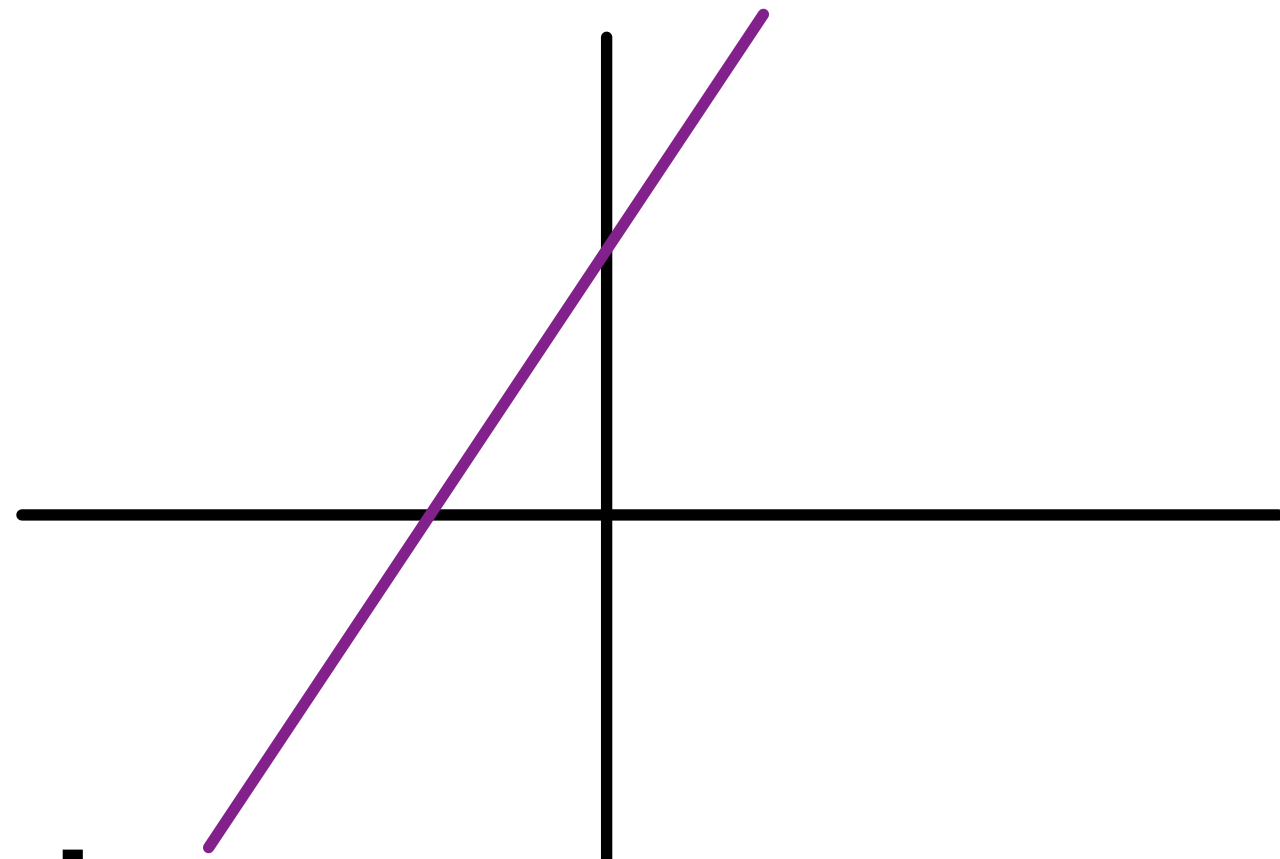
Does a system have a solution?

How many solutions are there?

What are its solutions?

Consistency

Consistency



Df. A system of linear equations is ***consistent*** if it has a solution

It is ***inconsistent*** if it has no solutions

demo

(consistency/inconsistency in $\mathbb{R}^3$)

Number of Solutions

Number of Solutions

zero the system is inconsistent

one the system has a unique solution

many the system has infinity solutions

Number of Solutions

zero the system is inconsistent

one the system has a unique solution

many the system has infinity solutions

These are the **only** options

Matrix Representations

$$2x + 3y = -8$$

$$y = 2$$

$$2y = 0$$

Matrix Representations

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix} \approx \begin{array}{l} 2x + 3y = -8 \\ y = 2 \\ 2y = 0 \end{array}$$

Matrix Representations

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix} \approx \begin{array}{l} 2x + 3y = -8 \\ y = 2 \\ 2y = 0 \end{array}$$

Writing down the unknowns is *tedious* (and more difficult to input into a computer)

Matrix Representations

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix} \approx \begin{array}{l} 2x + 3y = -8 \\ y = 2 \\ 2y = 0 \end{array}$$

Writing down the unknowns is *tedious* (and more difficult to input into a computer)

We'll write down linear systems as **matrices**, i.e., 2D grids of numbers with *fixed* width and height

Matrix Representations

$$\begin{bmatrix} 2 & 3 & -8 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix} \approx \begin{array}{l} 2x + 3y = -8 \\ y = 2 \\ 2y = 0 \end{array}$$

Writing down the unknowns is *tedious* (and more difficult to input into a computer)

We'll write down linear systems as **matrices**, i.e., 2D grids of numbers with *fixed* width and height

a matrix is just a representation

Matrix Representations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

Matrix Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

Matrix Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

Matrix Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

coefficient matrix

Matrix Representations

$$6\alpha - 2\beta - \gamma = 0 \quad (\text{C})$$

$$12\alpha - 6\beta = 0 \quad (\text{H})$$

$$6\alpha - \beta - 2\gamma = 0 \quad (\text{O})$$

Matrix Representations

$$\begin{bmatrix} 6 & -2 & -1 & 0 \\ 12 & -6 & 0 & 0 \\ 6 & -1 & -2 & 0 \end{bmatrix}$$

Solving Linear Systems

Solving Systems with Two Variables

$$(0, -2)$$

$$2(0) + 3(-2) = -6$$

$$4(0) - 5(-2) = 10$$

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

The Approach

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$2x = (-3)y - 6$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$4x - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$4((-3/2)y - 3) - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$-6y - 12 - 5y = 10$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$-11y = 22$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)y - 3$$

$$y = -2$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = (-3/2)(-2) - 3$$

$$y = -2$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = 3 - 3$$

$$y = -2$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Solving Systems with Two Variables

$$x = 0$$

$$y = -2$$

The Approach

Solve for x in terms of y in EQ1

Substitute result for x in EQ2 and solve for y

Substitute result for y in EQ1 and solve for x

Question

This works, but *why might we want a different approach?*

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Another Approach

Elimination

Eliminate x from the EQ2 and solve for y

Eliminate y from EQ1 and solve for x

Back-Substitution

Solving Systems with Two Variables

$$2x + 3y = -6$$

$$6x - 2y = 4$$

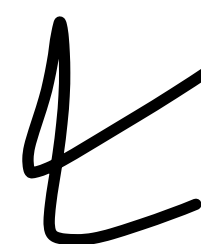
same solution

$$2x + 3y = -6$$

$$4x - 5y = 10$$

+

$$6x - 2y = 4$$



$$2x + 3y = -6$$

$$-11y = 22$$

$$4x - 5y = 10$$

$$- 2(2x + 3y) = 2(-6)$$

$$-11y = 22$$

$$y = -2$$

Motivating Question:
How does this look with matrices?

Solving Systems as Matrices

Observation. Each intermediate step of elimination and back-substitution gives us a new linear system with the *same solutions*

Solving Systems as Matrices

Observation. Each intermediate step of elimination and back-substitution gives us a new linear system with the *same solutions*

Can we represent these intermediate steps as operations on matrices?

Let's look back at this...

$$2x + 3y = -6$$

$$4x - 5y = 10$$

Elementary Row Operations

scaling	multiply a row by a NONZERO number
replacement	add a multiple of one row to another
interchange	switch two rows

Elementary Row Operations

scaling	multiply a row by a NONZERO number
replacement	add a multiple of one row to another
interchange	switch two rows

These operations don't change the solutions

Scaling Example

Scaling Example

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_1 \leftarrow 2R_1$$


$$\begin{array}{l} 4x + 6y = -12 \\ 4x - 5y = 10 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 6 & -12 \\ 4 & -5 & 10 \end{bmatrix}$$

Replacement Example

$$\begin{array}{l} 2x + 3y = -6 \\ 4x - 5y = 10 \end{array}$$

$$R_2 \leftarrow R_2 + R_1$$



$$\begin{array}{l} 2x + 3y = -6 \\ 6x - 2y = 4 \end{array}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 6 & -2 & 4 \end{bmatrix}$$

Replacement Example

Interchange Example

$$\begin{aligned} 2x + 3y &= -6 \\ 4x - 5y &= 10 \end{aligned}$$

$$R_1 \leftrightarrow R_2$$



$$\begin{aligned} 4x - 5y &= 10 \\ 2x + 3y &= -6 \end{aligned}$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -5 & 10 \\ 2 & 3 & -6 \end{bmatrix}$$

Interchange Example

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

$$R_2 \leftarrow R_2 / (-11)$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

$$R_2 \leftarrow R_2 / (-11)$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2$$



$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Example: Row Reductions

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & -11 & 22 \end{bmatrix}$$

$$R_2 \leftarrow R_2 / (-11)$$



$$\begin{bmatrix} 2 & 3 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 3R_2$$



$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 / 2$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Example: Row Reductions

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_2 \leftarrow R_2 / (-11)$$

$$R_1 \leftarrow R_1 - 3R_2$$

$$R_1 \leftarrow R_1 / 2$$

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Example: Row Reductions

$$R_2 \leftarrow R_2 - 2R_1$$

$$R_2 \leftarrow R_2 / (-11)$$

elimination

$$R_1 \leftarrow R_1 - 3R_2$$

$$R_1 \leftarrow R_1 / 2$$

back-substitution

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Row Equivalence

Df. Two matrices are *row equivalent* if one can be transformed into the other by a sequence of row operations

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Row Equivalence

Df. Two matrices are *row equivalent* if one can be transformed into the other by a sequence of row operations

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

We can compute solutions by sequence of row operations

Question

Question

How do we know when we're done? What is the "target" matrix?

Question

How do we know when we're done? What is the "target" matrix?

When the final matrix "looks like" a solution

Question

How do we know when we're done? What is the "target" matrix?

When the final matrix "looks like" a solution

We'll get more into that next time...

Example

$$x - 2y + z = 5$$

$$2y - 8z = -4$$

$$6x + 5y + 9z = -4$$

Summary

Linear equations define *hyperplanes*

Systems of linear equations may or may not have *solutions*

Linear systems can be represented as *matrices*, which makes them more convenient to solve